

Hao Yan

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Summary

Ph.D. candidate in Computer Science at George Mason University, advised by Dr. Ziyu Yao. My research focuses on natural language processing and large language models, with a unifying goal of trustworthy, reliable, and efficient LLMs.

Research Interests: LLM for Code Generation, LLM Safety and Trustworthiness, Agentic AI, Mechanistic Interpretability of LLMs, LLM Explainability, Human-AI Interaction, and Persona Simulation.

Education

George Mason University

Ph.D. in Computer Science, advised by Dr. Ziyu Yao

M.S. in Computer Science

Fairfax, VA

Aug. 2021 – Expected Dec. 2026

Aug. 2018 – Dec. 2020

Fujian Normal University

B.E. in Electronic and Information Engineering

Fujian, China

Aug. 2013 – Jun. 2017

Publications (* denotes equal contribution)

1. **Hao Yan**, Ziyu Yao. “Interpreting and Steering for Safe and Correct Code Generation”. **EMNLP 2026**, Long Paper. [link]
 - Mechanistic-interpretability study of how code LLMs represent safe vs. vulnerable generation; built CodeSec-Pairs, a dataset of 9,342 contrastive code pairs across 5 CWEs.
 - Proposed DuoSteer, a training-free inference-time steering method that cuts the vulnerability rate of generated code by 26.9% and improves functional correctness by 7.5%.
2. **Hao Yan**, Swapneel Suhas Vaidya, Xiaokuan Zhang, Ziyu Yao. “Guiding AI to Fix Its Own Flaws: An Empirical Study on LLM-Driven Secure Code Generation”. **ACM Transactions on Software Engineering and Methodology (TOSEM) 2026**. [link]
 - Large-scale empirical study of preventing and repairing security vulnerabilities in LLM-generated code across proprietary and open-weight models.
 - Found that directive self-generated vulnerability hints best prevent insecure code, and that detailed, actionable feedback best improves post-hoc repair.
3. Saaketh Anand Sai Kumar*, **Hao Yan***, Sai Charan Perepa, Murong Yue, Ziyu Yao. “Can LLMs Simulate Personas with Reversed Performance? A Systematic Investigation for Counterfactual Instruction Following in Math Reasoning Context”. **COLM 2025 Workshop** on Social Simulation with LLMs. [link]
 - Introduced the first benchmark for faithful counterfactual persona simulation with the Degree of Contrast (DoC) metric; showed that state-of-the-art models (e.g., o1) fail to faithfully simulate low-performing personas.
4. **Hao Yan**, Thomas D. LaToza, Ziyu Yao. “IntelliExplain: Enhancing Conversational Code Generation for Non-Professional Programmers”. **arXiv preprint**, 2024. [link]
 - Designed a new interaction paradigm with clearer explanations and a structured feedback loop that improves non-experts’ code comprehension and task efficiency in user studies.
5. **Hao Yan**, Saurabh Srivastava, Yintao Tai, Sida I. Wang, Wen-tau Yih, Ziyu Yao. “Learning to Simulate Natural Language Feedback for Interactive Semantic Parsing”. **ACL 2023**, Long Paper. [link]
 - Introduced the task of learning to simulate natural-language user feedback for training interactive semantic parsers; matched full-human-feedback performance on text-to-SQL even in low-data settings.
6. Saurabh Srivastava, Janit Bidhan, **Hao Yan**, et al. “Reasoning Under Constraint: How Batch Prompting Suppresses Overthinking in Reasoning Models”. **ICLR 2026 Workshop** on Logical Reasoning of Large Language Models. [link]
 - Showed that batch prompting suppresses “overthinking” in reasoning LLMs and cuts unnecessary reasoning steps for lower-cost inference without loss of answer quality.

Experience

Applied Scientist Intern, Amazon (Seattle, WA)

May 2026 – Aug. 2026

- Researched evolutionary strategies (ES) for backpropagation-free LLM post-training at production scale.
- Designed an improved evolutionary strategy with higher task performance and better training efficiency.

- Evaluated ES post-trained models against GRPO-trained baselines and analyzed performance–efficiency trade-offs.

Applied Scientist Intern, Amazon (*Seattle, WA*)

May 2025 – Aug. 2025

- Built a reward model that incorporates personalized user preferences into its evaluation of smart-home actions (API calls) and serves as both an automatic evaluator and the reward signal for GRPO-based post-training.
- Designed a distillation-based training strategy for the reward model and implemented the end-to-end data-processing and model-training pipeline.
- Evaluated state-of-the-art reward-modeling methods and led error analysis to identify future improvements.

Machine Learning Engineer Intern, Salesteq (*Remote*)

Jan. 2025 – Apr. 2025

- Participated in the design of an agentic LLM framework for car sales and service workflows, using AutoGen.
- Built conversation memory and RAG-style car-inventory retrieval to ground recommendations in dealership data.
- Integrated external tools (email, scheduling APIs) via structured prompting and adaptive control logic.

Graduate Research & Teaching Assistant, George Mason University (*Fairfax, VA*)

Aug. 2021 – Present

- Conduct research on LLM for Code Generation, LLM Safety and Trustworthiness, Agentic AI, Mechanistic Interpretability of LLMs, LLM Explainability, Human–AI Interaction, Persona Simulation, and Efficient LLM Reasoning.
- Authored multiple first-author publications on these topics.

Front-end Developer, Bright Sky Tech Inc. (*Annandale, VA*)

Jan. 2021 – Aug. 2021

- Built and maintained a web-based online grocery platform using Angular, Firebase, TypeScript, and Ionic.

Professional Service

Program Committees & Reviewing

- **Program Committee Member:** IJCAI 2026, 2025; ECAI 2025.
- **Reviewer:** NeurIPS 2026; ICLR 2026, 2025; ACL ARR 2026, 2025; IEEE CI-NLPSoMe 2025; WLLFM 2024.

Invited Talks

- “LLM Agents: Build Your Own Personal Assistant,” EPIC SPICE Intern Program, George Mason University, 2024.

Volunteer

- Student Volunteer, 61st Annual Meeting of the Association for Computational Linguistics (ACL), 2023.

Honors & Awards

- **CAHMP Summer GRA Award**, George Mason University (May 2024)
- **Graduate Student Travel Fund (GSTF)**, George Mason University (July 2023)
- **Distinguished GTA Award**, George Mason University (May 2022)
- **Graduate Students Poster Award (Honorable Mention)**, George Mason University (May 2022)
- **Outstanding GTA Award**, George Mason University (May 2020, May 2021)

Technical Skills

Programming: Python, SQL, Java, JavaScript, TypeScript.

ML & LLM Frameworks: PyTorch, Hugging Face (Transformers, Datasets, PEFT), vLLM, AutoGen, scikit-learn, NumPy, Pandas.

LLM Post-Training: Supervised Fine-Tuning & LoRA, RLHF, GRPO, Evolutionary Strategies, Reward Modeling, Knowledge Distillation.

LLM Systems & Agents: Agentic Frameworks, Tool Use, Retrieval-Augmented Generation, Prompt Engineering.

Evaluation & Research Methods: Benchmark & Dataset Construction, Model Evaluation, Error Analysis, Human & LLM-Based User Studies, Model Interpretability & Explainability.

Infrastructure: Linux, Git, AWS, Docker, GPU/Cluster Training, Weights & Biases.